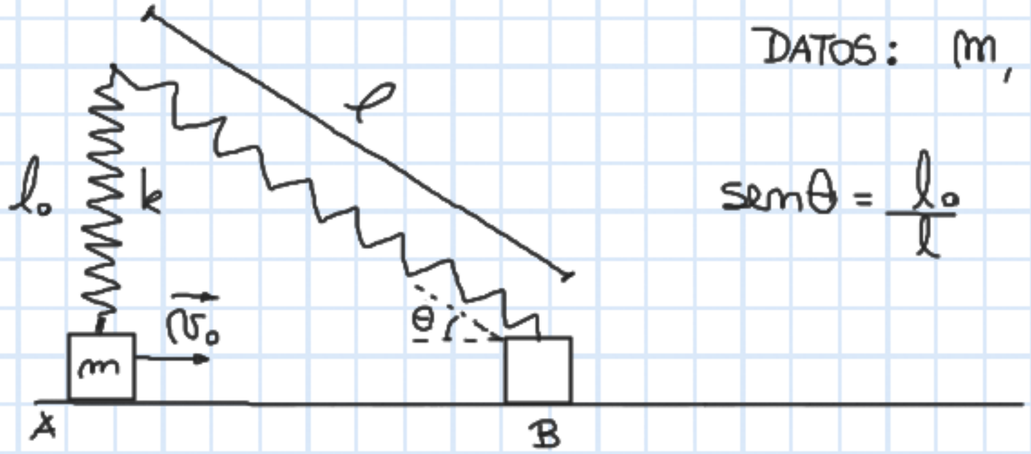
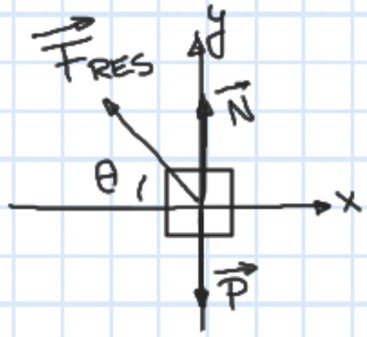


DATOS: m, g, k, l_0

$$\operatorname{sen}\theta = \frac{l_0}{l}$$



$$\Sigma F_y = N + F_{RES}\operatorname{sen}\theta - P = 0$$

$$N = P - F_{RES}\operatorname{sen}\theta = 0$$

$$F_{RES}\operatorname{sen}\theta = mg$$

$$k(l - l_0)\operatorname{sen}\theta = mg$$

$$k(l - l_0)\frac{l_0}{l} = mg$$

$$(l - l_0)\frac{l_0}{l} = \frac{mg}{k}$$

$$l - l_0 = \frac{mg}{kl_0}\cdot l$$

a)

$$\boxed{l = \frac{l_0}{1 - \frac{mg}{kl_0}}}$$

b)

$$E_{M_A} = E_{M_B}$$

$$\cancel{\tfrac{1}{2}} m v_A^2 = \cancel{\tfrac{1}{2}} k \Delta x^2$$

DONDE $\Delta x = l - l_0$

$v_A = v_0$

$$m v_0^2 = k \Delta x^2$$

$$v_0^2 = \frac{k}{m}(l - l_0)^2$$

$$v_0^2 = \frac{k}{m}\left[\frac{l_0}{1 - \frac{mg}{kl_0}} - l_0\right]^2$$

$$v_0^2 = \frac{k}{m} l_0^2 \left(\frac{kl_0}{kl_0 - mg} - 1\right)^2$$

$$\boxed{v_0 = l_0\sqrt{\frac{k}{m}}\left(\frac{mg}{kl_0 - mg}\right)}$$